Periodic Review as Sampled Governance: Sample-and-Hold Dynamics of Assessment-Driven Effort Control, a Selected 42-Stock Spectral Screen, and the Northern Cod Case

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

September 11, 2026

Abstract

Fisheries governance is periodic: assessments are compiled at fixed cadences, decisions taken at review points, and controls held until the next review. Real institutions therefore operate a sampled-control loop — observation and update occur at discrete review times — an architecture we call *sample-and-hold governance*. The formal literature instead represents institutional response as a continuous-time delay or an annual discrete step on a surplus-production model; neither matches the operating object, and substituting either can move or delete stability boundaries.

We analyse a logistic stock under held effort between reviews, with a filtered deficit signal carrying institutional memory and a projected Euler update resetting effort at each review. The sampled state space is forward invariant by induction; the rapid-review limit is a finite-horizon consistency statement, not a stability claim. The logistic hold map's equilibrium multipliers cross the unit circle at a review interval near 6.5 yr (a Neimark–Sacker-type crossing); thresholds reported by first-order discretisation are command-step artefacts. The archived stage-structured response regions are provisional (their generating computation was never attached) and not robust to the catchability scale the stage record never specifies: at $q = 0.1$ every class's annual-review verdict flips, making that comparison uninformative rather than a non-reproduction. The sample-and-hold map and the continuous-delay equation are distinct operators; stability does not transfer in general between them.

A multiplicity-controlled Lomb–Scargle screen of 42 annually assessed stocks finds no target-band discoveries; a structured search across more than thirty systems returns zero eligible cases. The northern cod case yields a descriptive split: the crash interpretation is formulation-dependent, post-collapse dynamics expose an identification problem the mortality-allocation comparison does not resolve, and a phase-line obstruction shows why no fixed scalar autonomous model reproduces the observed reversals.

The cadence and form of periodic review, not ecological lag alone, determine whether governance stabilises or destabilises a harvested stock; the review interval is a local spectral design parameter.

Keywords: periodic review; sample-and-hold governance; review interval; institutional delay; sampled-data control; Neimark–Sacker bifurcation; fisheries management

Box 1. Claims at their exact evidential status.

Each central claim, its evidential status, and its record pointer.

Claim	Evidential status	Record
Forward invariance of the sampled state space	Exact (induction)	Section 3.1, Proposition 3.1
Rapid-review limit: finite-horizon consistency only	Exact, scoped	Section 3.2
Small-T_r multiplier transfer $\mu_j = 1 + T_r\lambda_j + O(T_r^2)$	Conditional (hyperbolicity, C^1-consistency, projection inactive)	Section 3.2
Euler hold map: complex pair at 47.536 yr; real -1 at 79.143 yr; stable band [47.536, 79.143] yr	Closed-form monodromy, 200,001-point scan with bisection; command-step artefacts relative to the exact map	Section 3.4
Euler protective channel: real -1 at 2.306 yr, stable only on [0.2, 2.306] yr	Same record; artefact band	Section 3.4
Exact map: single complex crossing at 6.501 yr (unstable to stable)	Closed-form monodromy record	Section 3.4
Exact protective channel: no crossing; maximum $\rho = 0.9967$	Same record	Section 3.4
Annual review unstable: $\rho = 1.00035$ (exact) and 1.00055 (Euler); protective 0.9838	Multiplier record; near-unit-circle margins with sensitivity caveat	Section 3.4
Continuous-delay Hopf pair at 3.666 and 150.358 yr on the same plant	Companion study (Abaee, 2026)	Section 3.4
Relation between the three undelayed-limit statements	Reconciled conditional on the companion's undelayed sign ($\operatorname{Re}\lambda > 0$); λ itself unlisted here	Section 3.4
Stage-map archived windows: anchovy 3–4 yr, sprat 6–12 yr, cod convergence on [1, 20] yr, slow-stock oscillation-then-convergence	Archived, unreproduced (generating computation unattached)	Section 3.3
Screen bands 4–8 yr (biomass) and 12–60 yr (effort)	Descend from the archived stage peaks; inherit provisional status	Sections 2.4 and 3.3
Stage reconstruction: annual stability $\rho(1) = 0.895$ (anchovy), 0.923 (sprat), 0.956 (cod), 0.994 (slow-stock); long-horizon band from 34–35 yr; slow-stock $\rho(50) = 0.67$	Specified, pre-registered reconstruction (plan fixed before any run)	Section 3.4
Stage reconstruction at $q = 0.1$: every class unstable at annual review ($\rho(1) \geq 1.29$); slow-stock $\rho(50) = 7.8$	Sensitivity layer of the reconstruction	Section 3.4
Archived-versus-reconstruction verdicts (MATCH/MISMATCH table)	Post-hoc consistency check; uninformative at the unspecified q	Section 3.4
Screen of 42 annually assessed stocks: zero robust target-band peaks	BH-adjusted selected-cohort consistency check	Section 3.5
Power: sprat-class 1.0 and 0.24–0.58; anchovy-class 0.02–0.14	Conditional simulations	Section 3.6

Claim	Evidential status	Record
Structured case search: zero eligible cases among more than thirty systems	Curated inventory	Sections 2.6 and 3.7
Anchoveta–ENSO association: 3.7 yr peak, $r = +0.51$, era-split 1950–1984	Archived-data battery; association, not an identified mechanism	Section 3.7
Northern cod crash window 1991–1995 (Table 2)	Assessment record (DFO CSAS SAR 2016/026, NCAM M-shift)	Sections 2.7 and 3.8
Constrained-M quantities ($M = 0.46$, $F = 1.37$, unreported catch 257.8 kt yr^{-1}, ...)	Unreproduced hypotheses (open problems)	Section 3.8
Northern cod two-window split	Descriptive partition	Sections 3.8 and 4.3
Post-2015 production stall	Review record (Rose, 2026)	Section 3.8
Phase-line obstruction	Exact; model-class diagnostic (its two conditions unmet by the assessment record)	Sections 2.7 and 3.8
Five prospective designs	Specified, unexecuted; no registration identifier	Section 4.5
Distributive constraints	Stated, not operationalised	Section 4.6 and Appendix B

1 Introduction

Fisheries governance is periodic. Surveys are scheduled, assessments are produced at fixed cadences, advice is delivered at review points, and the resulting catch or effort controls are held in force until the next review. The review interval — the time between successive opportunities to revise the command — is sometimes a single year, sometimes a multi-year plan, sometimes a moratorium of indefinite length (Punt and Donovan, 2007; DFO, 2016). The formal literature treats this institutional clock in two ways, neither of which matches the operating object. Population-dynamic theory models institutional response either as a continuous-time lag, that is, a delayed signal entering an evolving control law, in the lineage of Gurney, Blythe, and Nisbet (1980); or as an annual discrete-time decision appended to a surplus-production model, in which the between-decision dynamics are compressed into a single annual step. Real institutions do neither. They observe and assess at discrete times, choose a rule-based command, and then hold or interpolate that command until the next decision opportunity. This paper calls that architecture *sampled governance* — a governance loop in which state observation and control update occur at discrete review times, not continuously. It is the same architecture the title and abstract call *sample-and-hold governance*: one object under two names — *sampled governance* names the institutional loop, *sample-and-hold* its control-theoretic update law — and the two names are used interchangeably throughout this article.

Two failure modes follow from this mismatch. The first is *architecture substitution.* Replacing the sample-and-hold architecture by a continuous delay, or by an annual difference equation, can move or delete stability boundaries. Conclusions about a governance system then depend on an operator that the institution does not implement. The second is *causal promotion.* Spectral peaks, cohort cycles, and crisis-driven monitoring records circulate as evidence for institutional-feedback mechanisms when they carry no information about the controller's sign, cadence, or timing at all. The marine-science literature already contains the disciplinary template for resisting this promotion. The skeptical re-analysis of the Russell Cycle (McManus, Licandro, and Coombs, 2016) showed that an 88-year zooplankton series long treated as a cycle failed to show a resolvable true periodicity. This article extends that discipline from climate-forced cycles to governance-feedback claims.

The contribution is fivefold. First, a sample-and-hold model of periodic review is specified, in which the governance clock is decomposed into seven time objects (observation interval, assessment interval, review interval, decision lag, deployment lag, ecological response lag, and memory timescale) and the extractive controller is an explicit discretisation with specified comparator classes. Second, two properties of the sampled process are established exactly: forward invariance of the sampled state space, and the precise scope of the rapid-review limit — a finite-horizon numerical-consistency statement that implies nothing about stability at positive review intervals. Third, the review-interval spectrum is computed under the discipline that the sample-and-hold map and the continuous-delay equation are *different operators*. A stability window located on one map does not transfer to the other, and the paper states which operator carries which statement. Fourth, an empirical layer is reported: a multiplicity-controlled spectral screen of 42 annually assessed stocks, a power analysis of that screen, and a structured case search across more than thirty resource systems. Fifth, the northern cod case (NAFO 2J3KL) is carried through the resulting falsification discipline. The paper's constructive content is the prospective programme — five identification designs and a closed-loop evaluation design specified as preregistration targets — that could convert or refute the mechanism.

The model class examined here is the extractive controller: a rule that raises extraction effort in response to a perceived decline. This is the demand-side form of the overshoot loop. The response to a perceived shortfall falls on the productive stock itself rather than on its regeneration. Extraction is raised precisely because the base has declined, which accelerates the decline it is reacting to. Its protective counterpart (effort reduced in response to decline), fixed multi-year plans, and hybrid rules serve as comparators. Conclusions drawn inside the extractive class are not generalised to governance as a whole, and the screen below asks whether such a channel is empirically resolvable at all. The implication for institutional design is direct: an annual-review extractive rule and a multi-year protective plan are not interchangeable instruments, and a sampled-data analysis must state which operator a given stability claim is computed on.

2 Material and methods

2.1 The sample-and-hold model of periodic review

The model treats periodic review as a sample-and-hold control architecture: the command set at a review instant is held fixed until the next review, and only the between-review ecological dynamics are continuous in time. Seven time objects are kept distinct, and no two are collapsed into a single "governance lag" unless the empirical record cannot resolve them and the aggregation rule is stated (Table 1).

Table 1. The governance-time ontology.

Object	Definition
Observation interval T_{obs}	Time between raw measurements
Assessment interval T_{assess}	Time between formal state estimates
Review interval T_r	Time between opportunities to change the command
Decision lag τ_{dec}	Assessment completion to formal decision
Deployment lag τ_{dep}	Decision to implemented change in extraction pressure
Ecological response lag τ_{eco}	Implementation to detectable ecological response
Memory timescale τ_m	Relaxation time of a filtered institutional signal; not a discrete delay

Reviews can be annual while deployment is delayed; observations can be frequent while decisions are legally fixed for several years; and stock and realised-effort series alone may identify only a combined closed-loop phase shift rather than the separate τ_{dec} and τ_{dep}. Separating them empirically requires dated observation releases, assessment products, commands, and implementation records. Alternatively, a proved structural-identifiability argument for the specified observation model can serve. Two further conventions apply. The review opportunity is not the same as the response sign: the extractive command studied here raises effort in response to a perceived decline, and the protective controller enters only as a comparator. The fixed-τ form is also an idealisation. Real institutions confronting visible scarcity typically change their instrument set — buyouts, engineered transfers, new infrastructure — rather than hold a single response law with constant delay. Any estimated τ therefore summarises a changing control architecture, not a structural constant.

Notation. Throughout, N denotes the resource stock, Z the institutional deficit signal, and E the extraction effort. The review Poincaré map — the discrete-time map obtained by sampling the continuous flow at review instants — is written $\mathcal{P}_{T_r}$; its Jacobian at the fixed point X^*, the monodromy matrix of the sampled loop, is $D\mathcal{P}_{T_r}(X^*)$ (Section 2.3's multiplier condition is written on it); the projection onto the admissible effort interval $[0, E_{\max}]$ is $\Pi_{[0,E_{\max}]}$. The signal map Φ is a nonnegative functional of the deficit, with memory timescale $\tau_m > 0$; its softplus realisation is Φ_k, with shift $\delta = \log 2/10 \approx 0.069$ and sharpness k (the shift is the equilibrium memory level, distinct from and unrelated to the effort-law gain δ_0, as the display below states). The effort law is F_B. The assessment operator is $\mathcal{A}_n$ and the observation operator is $\mathcal{O}$. In the control sections $S(\cdot)$ is surplus production — $S(N)$ on the logistic plant (defined after equation (2)) and $S(A)$ on the stage plant (Section 3.4) — while g (Section 3.3), C_E and C_Z (Section 3.4), and the four-state loop (Section 3.7) are defined where they are used. In the northern cod case (Sections 2.7 and 3.8, Table 2) S is instead spawning stock biomass (reported as SSB in Table 2), $\mathfrak{s}$ the unstable threshold, K the unexploited carrying capacity, $C(t)$ removals, M instantaneous natural mortality (with extra mortality M_x in Lemma 2.2), and F fishing mortality. The two scopes of S never share an equation, and no other symbol serves two sorts: the monodromy is written $D\mathcal{P}_{T_r}(X^*)$ throughout, and Lemma 2.2's production maximum is written $f_{\max}$, leaving M to natural mortality alone.

Let review times be $t_n = nT_r$. Between reviews, extraction effort is held at E_n and the resource stock follows the logistic process under held effort:

$$\dot{N}(t) = rN(t)\left(1 - \frac{N(t)}{K}\right) - qE_nN(t), \qquad t \in [t_n, t_{n+1}), \tag{1}$$

while the institutional signal Z is the filtered nonnegative deficit

$$\dot{Z}(t) = \frac{\Phi\big(qE_nN(t) - S(N(t))\big) - Z(t)}{\tau_m}, \qquad \tau_m > 0, \tag{2}$$

with $S(N) = rN(1 - N/K)$ the surplus production and $\Phi \geq 0$ a nonnegative signal map with memory timescale τ_m. The assessment available at review n is

$$\widehat{Z}_n = \mathcal{A}_n(\{Y_j : t_j \leq t_n - \tau_{\mathrm{dec}}\}), \qquad Y_j = \mathcal{O}\big(N(t_j)\big) + \varepsilon_j, \tag{3}$$

which separates the latent process N, the observations Y, and the assessments $\widehat{Z}$. Measurement and assessment errors need not be independent or Gaussian, but the admissible error support is restricted to $\widehat{Z}_n \geq 0$. The multiplicative-error robustness experiment of Section 3.3 satisfies this by construction. In the executed calibration all parameters entering the effort law are positive (r, K, q, $E_{\max}$, Δ_{ref}, Z_{ref}, $\tau_m > 0$); because $Z_{\mathrm{ref}} > 0$ and $\widehat{Z}_n \geq 0$, the rational term is well defined on the admissible state space. The effort law's rational term is undefined at $\widehat{Z}_n = -Z_{\mathrm{ref}}$ and changes sign beyond it, and projection of the final command does not repair an undefined update. In the baseline deterministic record the assessment operator is exact and contemporaneous, $\widehat{Z}_n = Z(t_n^-)$: each command uses the just-flowed end-of-period state, indexed by the review that reads it ($\widehat{Z}_{n+1}$ in (4)); the exact update holds this same flowed value. The review map is the projected forward-Euler step

$$E_{n+1}^{\mathrm{cmd}} = \Pi_{[0,E_{\max}]}\left\{E_n + T_r F_B(E_n, \widehat{Z}_{n+1})\right\}, \tag{4}$$

— one specific reviewed controller among many (direct command rules, exact integration under held assessment, incremental rules with fixed step). Because the increment scales with T_r, a sweep in T_r changes both the hold duration and the accumulated gain per review. The separating computation is the exact held-assessment update (the exact update), reported in Section 3.4 with attribution to the companion delay study (Abaee, 2026). The effort law is

$$F_B(E,Z) = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z}{\Delta_{\mathrm{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0 \frac{Z}{Z_{\mathrm{ref}} + Z}\right],$$

The protective comparator is the quota-tracking law

$$F_B^{\mathrm{prot}}(E,Z) = \left(1 - \frac{E}{E_{\max}}\right)\eta_p\big(E_{\mathrm{cap}}(Z) - E\big), \qquad E_{\mathrm{cap}}(Z) = E_0 \frac{Z_{\mathrm{ref}}}{Z_{\mathrm{ref}} + Z},$$

with $\eta_p = \eta$ and $E_0 = E^*(Z_{\mathrm{ref}} + \delta)/Z_{\mathrm{ref}}$, calibrated so its unique interior rest coincides with the extractive fixed point; its linearised gains there are $C_E = -0.850336$, $C_Z = -1.661702$ (Abaee, 2026, equation (3)), reviewed through the same projection $\Pi_{[0,E_{\max}]}$.

together with the shifted, floored softplus signal map

$$\Phi_k(s) = \max\left\{0,\ \frac{1}{k}\log\!\big(1 + e^{ks}\big) - \frac{\log 2}{k} + \delta\right\}.$$

with shift $\delta = \log 2/10 \approx 0.069$, the equilibrium memory level ($Z^* = \Phi(0) = \delta$; distinct from and unrelated to the effort-law gain δ_0) and sharpness k — the value used in the computed records, $k = 10$, is stated where it is used in Section 3.4 and collected in Table 3 of Appendix A. The multiplier records of Section 3.4 presuppose a fixed point interior to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$; the equilibrium coordinates are not listed here (Appendix A).

Equations (1)–(4) with $\Phi = \Phi_k$ constitute the extractive controller studied throughout. It is an explicit controller discretisation, not a generic harvest-control rule and not a model of protective quota reduction. Three comparators are distinctly specified: the protective controller (the quota-tracking law above, the same machinery with the response to decline entering with the opposite sign), the fixed-plan controller (no state-responsive update between scheduled resets), and the hybrid controller (annual-change limits, emergency triggers, or legal overrides). The comparators are specified for the prospective management-strategy evaluation (Section 4.5), not presented as completed experiments. A fixed plan and a state-responsive plan with the same nominal review period are not the same intervention. The model-level run of the comparator evaluation on the logistic hold-map core — the plant of the closed-form operator results — is executed and reported in Section 3.4 at the stated evidential status (deterministic, seed-fixed, on the specified architecture only). The prospective real-system design of Section 4.5 remains unexecuted.

2.2 Registration conventions

The deterministic record fixes a one-step flow-then-update convention, stated in pre-review/post-review terms. On $[t_n, t_{n+1})$ effort is held at E_n. At the review instant t_{n+1} the contemporaneous, exact assessment $\widehat{Z}_{n+1} = Z(t_{n+1}^-)$ of the flowed state is read ($\tau_{\mathrm{dec}} = \tau_{\mathrm{dep}} = 0$), and equation (4) computes the command E_{n+1} held on the following interval. The pre-review state is $X_n^- = (N(t_n^-), Z(t_n^-), E_{n-1})$ and the map used is the pre-review Poincaré map $X_{n+1}^- = \mathcal{P}_{T_r}(X_n^-)$. The reversed ordering (command first, flow second) is a different hybrid system and is not used. No decision or deployment queue is present, and multiplicative assessment error enters only the designated robustness experiment. A positive τ_{dep} requires an explicitly specified hold or interpolation rule.

The computational status of the response-region records is stated with the same care. Classification used long-horizon trajectories, tail amplitudes, multiple histories, and integration-step refinement. It did not use Poincaré-map multipliers, monodromy matrices, or Floquet multipliers. The reported bands are therefore finite-grid, trajectory-classified response regions, and no Neimark–Sacker, flip, or other multiplier-crossing classification is claimed. Until the solver configuration and initial histories are attached (Appendix A), the stage-output values carry provisional status. For the logistic hold-map core that boundary calculation is reported in Section 3.4. For the stage-structured map the boundary record is supplied by the labelled reconstruction of Section 3.4 — a newly specified, pre-registered object whose full specification (flow/update ordering, information pattern, derivative construction, multiplier trajectories, nonlinear trajectories, solver configuration) is given there. The original stage record remains unattached, and its output values keep provisional status. The computational requirements of this section and of Sections 2.4, 2.5, 3.3, and 3.4 — solver configurations, initial histories, stock identifiers, the eligibility table, the null-calibration record, simulation code and seeds, and the reconstruction's plan, code, and output tables — are consolidated in Appendix A, which also fixes the pre-registration convention.

2.3 The review-map operators

Three objects are in play: the continuous-delay equation, the logistic hold map, and the stage-structured review map. The continuous delay equation and the logistic sample-and-hold map are the same feedback loop under two delay operators. The stage-structured map is a different ecological plant (delayed-recruitment stage structure against scalar logistic surplus production) with a different state dimension; it is not an operator substitution on the logistic plant. A valid architecture comparison holds the ecological plant, controller linearisation, equilibrium, and parameter vector fixed while changing only the timing operator. The 3–4 yr windows and the ≈ 6.5 yr exact-update crossing below are therefore reported as a comparison across plants and operators, which does not isolate the operator effect; the one-plant operator contrast in Section 3.4 isolates the operator for the logistic plant. Changing the operator can move or delete a crossing, and that relocation is a property of the review map's spectrum, not a refutation of the loop mechanism. The discipline throughout is that every stability statement names its operator, and statements computed on the hold map, on the stage-structured review map, and on the continuous-delay equation do not transfer to one another. The operative condition is $\det(D\mathcal{P}_{T_r}(X^*) - e^{i\theta}I) = 0$ on the map to which the statement refers, with $D\mathcal{P}_{T_r}(X^*)$ the Jacobian of the review map at its fixed point (the monodromy matrix of the sampled loop).

For a deterministic sampled system, the inter-review flow and the review update combine into a Poincaré map $X_{n+1} = \mathcal{P}_{T_r}(X_n)$, where X_n includes the ecological state, memory, held command, and any explicitly modelled queue. A fixed point is locally asymptotically stable if and only if every eigenvalue of $D\mathcal{P}_{T_r}(X^*)$ lies inside the unit disk. A verified crossing through the unit circle as the fixed parameter T_r changes is a sampled-data parameter bifurcation. It is not rate-induced tipping, which requires a nonautonomous parameter path and a loss of tracking as the ramp speed changes (Ashwin, Wieczorek, Vitolo, and Cox, 2012).

2.4 The cross-sectional spectral screen

The screen's input layer is a fixed 42-stock cohort of the RAM Legacy Stock Assessment Database v4.66 (Ricard, Minto, Jensen, and Baum, 2012), selected by a separate annual-review eligibility criterion. The release records stock and assessment metadata and series such as biomass, fishing mortality or exploitation rate, total allowable catch, catch advice, and effort. It does not encode controller sign, the decision rule, or dated decision and deployment queues. The annual-review designation is therefore a criterion applied in this analysis, not a database classification of a stock as extractive or protective.

For each eligible biomass and exploitation/effort proxy series, the analysis detrends according to a stated rule, computes a Lomb–Scargle periodogram (Lomb, 1976; Scargle, 1982), and integrates power in the prespecified bands (4–8 yr for biomass, 12–60 yr for effort) — bands derived from the archived, unreproduced stage-map diagnostics of Section 3.3 (its observable-specific dominant peaks near 4 and 8 yr in biomass and 12 and 60 yr in effort) and carry that record's provisional status into the screen's target definition. The tested statistic is band-integrated power, and a peak is separately classified for robustness. The result is compared with a per-series AR(1) red-noise null. P-values are adjusted across the specified family (stocks, observables, and bands) by the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995), which controls the false discovery rate, not the familywise error rate. Dependence across stocks and observables (shared climate forcing, shared assessment methods) is acknowledged, and the Benjamini–Yekutieli procedure under arbitrary dependence (Benjamini and Yekutieli, 2001) is the prespecified fallback. The reported zero count is the BH-adjusted result. One caveat: the 12–60 yr band is poorly resolved on records shorter than about three candidate periods, and at the long-period end band power is partly a trend test. A peak is classified as robust only if it survives the null comparison, the multiplicity adjustment, and sensitivity to detrending and endpoint choices.

2.5 Power analysis

Power experiments inject the model-generated effort signal into AR(1)-type noise and apply the same band-power statistic (the conventional power-analysis framing of Cohen, 1988). Power is estimated on 100–200 yr synthetic records across noise scales.

2.6 The structured case search

The case search is a curated inventory, not a systematic review. It considered more than thirty systems spanning fisheries, aquaculture, groundwater, surface water, rangeland, wildlife harvest, forestry, and produced-capital markets, under four criteria:

(i) a responsive institutional feedback rather than a one-time cap or ban;
(ii) an independently dateable response or implementation lag;
(iii) no major environmental, biological, or structural driver empirically supported as generating the same outcome, where the comparative rule is that a candidate institutional mechanism must outperform pre-registered alternatives on held-out prediction, phase ordering, or intervention response (since "capable of generating" is not a falsifiable exclusion in open ecological systems);
(iv) individual-resource or individual-station data rather than an aggregate series.

Behind criterion (iii) is an explicit list of the alternative mechanisms that can mimic or replace an institutional cycle: stage/maturation delays and cohort resonance; recruitment suppression versus stock culling; support-pool/nutrient limitation; pollution/waste feedback; predator–prey or climatic forcing; and direct abiotic mining and slow-store exhaustion. The rule is that a candidate institutional mechanism must be compared with these alternatives rather than identified from periodicity alone.

2.7 The northern cod case: constitutive model and evidence

The northern cod case (NAFO 2J3KL) is treated as a bounded empirical object. Its illustrative constitutive model is the strong-Allee surplus equation:

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right)\frac{S - \mathfrak{s}}{K - \mathfrak{s}} - C(t), \tag{5}$$

with S the spawning stock biomass, r the intrinsic growth rate, K the unexploited carrying capacity, $\mathfrak{s}$ the unstable threshold, and $C(t)$ the removals. The Schaefer model is the formal limit of this family ($\mathfrak{s} \to -\infty$) in which the factor $(S-\mathfrak{s})/(K-\mathfrak{s})$ is replaced by 1. No value of $\mathfrak{s}$ makes the displayed factor identically 1 (it tends to 1 only in the limit $\mathfrak{s} \to -\infty$), so the Schaefer comparison is a change of growth law, not a parameter specialisation. The constitutive equation is illustrative, not a fit — and the obstruction class is the one-dimensional autonomous equation with fixed parameters and fixed removals. The displayed equation is autonomous exactly when removals are fixed.

Proposition 2.1 (Phase-line obstruction). Let $f : \mathbb{R} \to \mathbb{R}$ be locally Lipschitz, and let $x(\cdot)$ be a nonconstant solution of the scalar autonomous ODE $\dot{x} = f(x)$. Then x is strictly monotone; it never attains an equilibrium value and cannot cross an equilibrium in either direction. Consequently, an exact path that repeatedly rises and falls across a common interval is incompatible with any single fixed scalar autonomous model.

Proof. If x is nonconstant on an interval and $x(t_1) = x(t_2)$ with $t_1 < t_2$, then by Rolle's theorem there exists $\tau \in (t_1, t_2)$ with $\dot{x}(\tau) = 0$, i.e. $f(x(\tau)) = 0$. By uniqueness the solution through $(\tau, x(\tau))$ is the constant equilibrium solution, a contradiction. Hence every solution is strictly monotone or constant. If a trajectory met an equilibrium value x^* at any time, it would be constant thereafter and, by uniqueness applied backwards, before; a nonconstant solution therefore never attains an equilibrium value and cannot cross one in either direction. A path that rises and falls across a common interval attains some level twice with opposite directions, contradicting monotonicity. □

Lemma 2.2 (Extra-loss threshold shift, conditional).

Statement. (i) Constant loss. In the constitutive model with an additional constant removal $C > 0$, the production function is replaced by $f_C(S) = f(S) - C$; the positive equilibria of the modified function are the solutions of $f(S) = C$, the smaller one (the effective threshold) lies to the right of $\mathfrak{s}$, and at the production maximum the two positive equilibria coalesce and beyond it disappear — all conditional on the solutions existing, i.e. on C being below the production maximum. For $C = 0$ the smaller positive equilibrium is $\mathfrak{s}$; the strict rightward shift holds for $C > 0$.

(ii) Proportional mortality. With extra mortality $M_x > 0$ the modified production is $f_M(S) = f(S) - M_x S$; positive equilibria solve $f(S)/S = M_x$, $S = 0$ remains an equilibrium (unlike case (i), where $\dot{S}|_0 = -C < 0$ drives the model below zero unless the removal is constrained near zero or the state equation is modified), and the same threshold-shift reading applies to the modified positive roots.

Case (i) subtracts a fixed term; case (ii) subtracts a term proportional to stock. In each case the smaller positive root of the modified production function moves rightward; at the production maximum the two positive equilibria coalesce, and beyond it they disappear. An effective threshold is not automatically a shifted structural parameter, and the coalescence/disappearance case is part of the statement.

Proof. The constitutive production $f(S) = rS(1 - S/K)(S - \mathfrak{s})/(K - \mathfrak{s})$ vanishes exactly at 0, $\mathfrak{s}$, K; it is negative on $(0, \mathfrak{s}) \cup (K, \infty)$, positive on $(\mathfrak{s}, K)$, and its derivative vanishes exactly at

$$S_\pm = \frac{K + \mathfrak{s} \pm \sqrt{K^2 - \mathfrak{s}K + \mathfrak{s}^2}}{3}, \qquad 0 < S_- < \mathfrak{s} < S_+ < K,$$

so f has exactly one critical point in $(\mathfrak{s}, K)$ — the unique maximum $S^* = S_+$ with value $f_{\max} = f(S^*) > 0$ — and f is strictly increasing on $(\mathfrak{s}, S^*)$ and strictly decreasing on (S^*, K).

(i) The equilibria of $f_C(S) = f(S) - C$ solve $f(S) = C$, so none lie outside $(\mathfrak{s}, K)$, where $f \le 0 < C$. For $0 < C < f_{\max}$, the intermediate value theorem on $[\mathfrak{s}, S^*]$ ($f(\mathfrak{s}) = 0 < C < f_{\max} = f(S^*)$) and on $[S^*, K]$ ($f(K) = 0 < C < f_{\max}$) supplies one root in each interval, and strict monotonicity makes each root unique. The smaller root satisfies $S_1(C) \in (\mathfrak{s}, S^*)$ — the effective threshold lies strictly to the right of $\mathfrak{s}$. As $C \uparrow f_{\max}$ the two roots approach S^* from either side (coalescence), and for $C > f_{\max}$ the equation $f(S) = C$ has no solution: the two positive equilibria have disappeared.

(ii) Write $f_M(S) = f(S) - M_x S = S\,(h(S) - M_x)$ with $h(S) = f(S)/S = \frac{r}{K-\mathfrak{s}}(1 - S/K)(S - \mathfrak{s})$, a quadratic with roots $\mathfrak{s}$ and K, negative on $(0, \mathfrak{s}) \cup (K, \infty)$, and unique maximum $H = h(S_h)$ at the vertex $S_h = (\mathfrak{s} + K)/2$. Then $f_M(0) = 0$: zero remains an equilibrium in every case. The positive equilibria solve $h(S) = M_x$; for $0 < M_x < H$ the same two-interval argument gives exactly two roots, the smaller in $(\mathfrak{s}, S_h)$ — the threshold shift — with coalescence at $M_x = H$ and disappearance for $M_x > H$, where $f_M(S) = S(h(S) - M_x) < 0$ on $(0, \infty)$.

In case (i), by contrast, $f_C(0) = -C < 0$, so $\dot{S}|_0 = -C$ drives the state below zero unless the removal is constrained near zero or the state equation is modified — as claimed. □

The case evidence is the assessment record (the recovery-potential and sequential stock assessments: DFO, 2011, 2016, 2022, 2024). The crash window (1991–1995) and the subsequent windows are documented by the assessment table of DFO CSAS SAR 2016/026 (Table A2), produced by the Northern Cod Assessment Model (NCAM; Cadigan, 2016), together with the dated governance events (the moratorium announcement of 2 July 1992; the reopening of 26 June 2024 with a total allowable catch of 18 kt). The displayed values are rounded values from the assessment table, and the survival column $\exp(-M)$ is a transformation of the reported instantaneous mortality estimate, not an independently observed survival series.

3 Results

3.1 Forward invariance of the sampled process

A first well-posedness check on the sampled controller is that the state space on which the sample-and-hold dynamics act — nonnegative stock, nonnegative memory signal, admissible effort — is preserved by the review map. This check comes first because every later result relies on the state remaining inside this set between reviews.

Proposition 3.1 (Forward invariance of the sampled process). Assume:

(H1) $N(0) \geq 0$, $Z(0) \geq 0$, $E_0 \in [0, E_{\max}]$.

(H2) The signal map Φ is non-negative.

(H3) Every effort command is projected by $\Pi_{[0,E_{\max}]}$.

(H4) Any between-review deployment interpolation remains in the convex interval joining consecutive commands. The main model takes $E(t) = E_n$, which satisfies (H4) trivially; the proposition is stated for the hold-or-interpolate class.

Then $N(t) \geq 0$, $Z(t) \geq 0$, and $E(t) \in [0, E_{\max}]$ for every time at which the sampled solution exists.

Proof. Proceed by induction over review intervals. Suppose that at a review time t_n one has $N(t_n) \geq 0$, $Z(t_n) \geq 0$, and that the command to be held or interpolated lies in $[0, E_{\max}]$. Projection places the next command in the same closed interval; a hold retains an endpoint of that interval, and any convex interpolation $E(t) = (1 - \theta(t))E_n + \theta(t)E_{n+1}$ with $0 \leq \theta(t) \leq 1$ also remains in $[0, E_{\max}]$. On $[t_n, t_{n+1}]$, Eq. (1) gives

$$N(t) = N(t_n) \exp\left\{ \int_{t_n}^{t} \left[r\left(1 - \frac{N(s)}{K}\right) - qE(s) \right] ds \right\} > 0$$

for a positive initial value, while if $N(t_n) = 0$, uniqueness gives the identically zero solution on the interval; N cannot become negative. Set $\nu(t) = \Phi(qE(t)N(t) - S(N(t))) \geq 0$. Variation of constants in Eq. (2) with $\tau_m > 0$ gives

$$Z(t) = e^{-(t-t_n)/\tau_m} Z(t_n) + \frac{1}{\tau_m} \int_{t_n}^{t} e^{-(t-s)/\tau_m} \nu(s)\, ds \ \geq 0.$$

Boundedness from above holds automatically on the stock equation and is used in the induction: on $[t_n, t_{n+1})$ the stock solves the logistic equation under held effort, and $\dot{N} \leq 0$ at $N = K$ (the harvest term is non-negative), so $N(t) \leq \max\{N(t_n), K\}$ throughout the interval. All three inequalities therefore hold through t_{n+1}; they hold at $t_0 = 0$ by hypothesis, and induction proves them on every review interval for which the sampled solution exists. □

Positivity, boundedness, persistence above a threshold, and viability are distinct properties; Proposition 3.1 establishes positivity and effort admissibility only. The institutional implication is modest but essential: the projected Euler step does not push the state outside its specified admissible set on a single review interval, and any later stability or instability claim is therefore a statement about dynamics inside that set, not about escape from it.

3.2 The rapid-review limit and what it does not establish

Define $u = (N, Z, E)$ and the continuous no-delay system $\dot{u} = G(u)$ assembled from Eqs. (1)–(2) with the effort law evaluated continuously. Suppose G is continuously differentiable with bounded derivative on a compact neighbourhood containing the compared trajectories on $[0, T]$, assessments equal the contemporaneous state, no additional decision or deployment queue is present, projection is inactive, and the sampled and continuous systems share the initial state. The frozen-effort flow followed by the explicit effort step then has a one-step defect of order $O(T_r^2)$ relative to the exact flow of G. The usual discrete Gronwall estimate yields an $O(T_r)$ review-time error and uniform convergence on every fixed finite horizon as $T_r \to 0$ (the sampled-data approximation framework of Nešić and Teel, 2004).

This is a numerical-approximation property, not a finite-review stability theorem. It gives no uniform-in-time estimate, does not preserve stability automatically, and does not control any trajectory-classified response region reported below. It excludes active projection, delayed or erroneous assessment, a deployment queue, and stochastic updates. In particular, it neither identifies a finite T_r with a continuous discrete delay nor implies that sufficiently small positive T_r is stable when the continuous target is unstable. The converse caution is equally important. Under the additional assumptions of hyperbolicity of the continuous equilibrium and C^1-consistency of the sampled map with projection inactive locally, the multipliers satisfy $\mu_j(T_r) = 1 + T_r \lambda_j(A) + O(T_r^2)$, so local stability persists for all sufficiently small positive review intervals provided every continuous eigenvalue satisfies $\Re\lambda_j < 0$; if any $\Re\lambda_j > 0$, the sampled equilibrium inherits instability for sufficiently small T_r. The finite-horizon result therefore neither proves nor refutes stability transfer; the transfer hypotheses are what any such claim would have to supply. The institutional reading is that "rapid review" is not a stability guarantee — a governance loop that reviews frequently approximates a continuous controller only on finite horizons, and the question of whether short review intervals stabilise or destabilise the equilibrium is decided by the spectrum of the map, not by the rapid-review limit.

3.3 Response regions of the delayed-recruitment review maps

The delayed-recruitment records, classified from long-horizon trajectories on the stage-structured review map, locate exploratory response regions by stock class (the delayed-recruitment oscillation literature of Gurney, Blythe, and Nisbet, 1980, supplies the classical reference point). These are archived records whose generating computation was never attached (Section 2.2): they are reported below at provisional status, and the pre-registered stage-map reconstruction of Section 3.4 does **not** reproduce the multi-year windows on the anchovy- or sprat-class parameterisation of that plant (reporting convergence there, with a separate long-horizon band that has no archived counterpart). The regions below are therefore archived, unreproduced records, not results of the reconstructed object:

- **Anchovy-class:** persistent tail oscillation near $T_r \approx 3$–4 yr, with a weak response at $T_r = 2$ yr; annual-review trajectories converge over the tested grids for every tested effort-response value.
- **Sprat-class:** persistent tail oscillation near $T_r \approx 6$–12 yr.
- **Cod-class:** trajectories converge to equilibrium for every tested $T_r \in [1, 20]$ yr, although the corresponding continuous-delay calculation has an oscillatory interval — a convergence-over-tested-grid record, not a stability theorem for every history.
- **Slow-stock class:** for $r \in (0.01, 0.05)$ yr^{-1}, oscillation over part of the tested grid below approximately 20–30 yr and convergence at longer review intervals, with transition brackets between approximately 30 and 50 yr depending on r. This one-sided pattern does not contradict rapid-review consistency: the continuous no-delay target is itself unstable over part of the slow-r regime, so small-T_r trajectories may approximate an unstable target on finite horizons. The record is conditional on the delayed-recruitment variant and is not a general claim that slower review stabilises governance; the dominant timescales are centuries (the archived slow-stock cohort cycle: $P \approx 250-360$ yr), beyond the length needed to resolve multiple cycles in most institutional records.

The corresponding continuous-delay calculations on the delayed-recruitment parameterisation — g is its maturation delay, and the response regions are located through the product rg — locate response regions near $rg \approx 1.5$–1.6: for $g = 2$ yr, $r \in (0.77, 0.81)$ yr^{-1} at $\eta = 0.914$ with a delay interval of approximately 2.6–7.8 yr; for $g = 1$ yr, the high-r interval is approximately 1.565–1.585 yr^{-1} with a delay interval of 1.6–3.5 yr. These are finite-grid trajectory summaries, not closed analytical stability regions.

Multiplicative assessment-error experiments preserve the anchovy-class trajectory region through 30% error and produce no noise-induced persistent tail oscillation at annual review in the tested ensemble. This is a robustness summary for the specified multiplicative perturbation only. The archived diagnostics carry observable-specific dominant peaks — approximately 4 yr in anchovy-class biomass with 12 yr in effort, and approximately 8 yr in sprat-class biomass with 60 yr in effort. These are reported only as observable-specific dominant peaks over the analysed windows: components of one stationary periodic orbit cannot have different fundamental periods, and harmonics, subharmonics, modulation, or transient spectral content are unresolved; no decomposition has established whether the baseline term, signal regularisation, or another controller component dominates these responses. The archived amplitudes indicate percent-scale biomass excursions and order-one effort excursions relative to equilibrium. The exact percentages are not used as effect-size estimates, because the archived record does not distinguish peak-to-peak range from half-range and the approach to the large effort response contains a long transient.

3.4 The operator spectra

The logistic hold-map core permits a closed-form expression for the held logistic flow; the linearised review map is then constructed from this flow, the signal dynamics, and the update derivatives. With effort held at E, the logistic flow over one review interval is, for $a(E) = r - qE \neq 0$,

$$N(t_{n+1}^-) = \frac{a(E)\, N(t_n^-)\, e^{a(E)T_r}}{a(E) + \frac{r}{K} N(t_n^-)\big(e^{a(E)T_r} - 1\big)},$$

and $N(t_n^-)/\big(1 + \frac{r}{K} N(t_n^-) T_r\big)$ for $a(E) = 0$. In computations, the limit branch is used when $|a(E)| < 10^{-12}$. The signal flow enters through equation (2) with this forcing, and the linearised pre-review monodromy is assembled from the held-flow state transition, the assessment derivative at $Z(t_{n+1}^-)$, and the update derivative of equation (4). The exact held-assessment controller — the separating comparator that isolates the Euler step — is the exponential update (the exact update) $e_{n+1} = e^{C_E T_r} e_n + \frac{e^{C_E T_r} - 1}{C_E} C_Z z_n$ for $C_E \neq 0$, and $e_{n+1} = e_n + T_r C_Z z_n$ for $C_E = 0$, with $e_n = E_n - E^*$ and $z_n = Z_n - Z^*$ the deviations from the compared fixed point — the exact solution, over one review interval, of the linearised effort law $\dot{e} = C_E e + C_Z z$ with the flowed end-of-period assessment z held, where C_E and C_Z are the coefficients of that linearisation (the partial derivatives of F_B with respect to E and Z at the compared fixed point; unrelated to the removals C of the cod case), and equation (4)'s increment is the forward-Euler discretisation of the same linear object. This comparator is analytical, not a proposed institutional rule. Comparing its monodromy with the forward-Euler map's separates review cadence, zero-order holding, and Euler discretisation. That comparison is reported here: executed and verified on the companion delay study's identical hold map (Abaee, 2026; same effort-law bracket, same softplus signal, same forward-Euler monodromy reproducing the 47.536 yr crossing reported here), the exact-update annual spectral radius is $\rho = 1.00035$ (Euler 1.00055); the 47.536 yr and 79.143 yr crossings are command-step artefacts of the forward-Euler update; the exact map's single unit-circle crossing is ≈ 6.5 yr; and the restabilising direction is preserved.

On the logistic hold-map core, the undelayed equilibrium is already unstable, so annual review is unstable. On the computed crossing set, the sampled equilibrium has a complex unit-circle crossing — the spectral signature of a Neimark–Sacker bifurcation, nonlinear non-degeneracy not verified — at $T_r^{\mathrm{UC}} = 47.536$ yr under the forward-Euler command step and at ≈ 6.5 yr under the exact update, with the same restabilising direction. The 47.536 yr crossing and its -1 multiplier at 79.143 yr are command-step artefacts, not a review-cadence property (Section 4.1). For an institution implementing the incremental Euler rule these crossings are dynamically real; they are artefacts only relative to the exact update.

The complete crossing record. On the logistic hold-map core the unit-circle crossing record over $[0.2, 200]$ yr is complete (Figure 1). It is computed from the linearised pre-review monodromy on a 200,001-point scan with bisection refinement (deterministic; it reproduces the numbers reported here on the range they cover). The forward-Euler update crosses twice on the extractive channel — a complex pair at $T_r = 47.536$ yr (unstable $\rightarrow$ stable) and a real -1 multiplier at 79.143 yr (stable $\rightarrow$ unstable), stable band $[47.536, 79.143]$ yr — and once on the protective channel — a real -1 at 2.306 yr (stable $\rightarrow$ unstable), stable only on $[0.2, 2.306]$ yr. The exact update crosses once on the extractive channel — a complex pair at 6.501 yr (unstable $\rightarrow$ stable), stable on $[6.501, 200]$ yr — and never on the protective channel, which is stable at every tested interval with maximum spectral radius 0.9967. Both Euler crossings of the extractive channel are therefore command-step artefacts relative to the exact map's single crossing, and the Euler protective instability beyond 2.306 yr is likewise an artefact: the exact protective map is stable throughout.

Spectral margins of the annual-review verdict. The annual-review instability is a near-unit-circle record, and its margins are part of the record. The exact update's annual spectral radius is $\rho = 1.00035$ — a modulus exceeding unity by 3.5×10^{-4} — with the forward-Euler update at 1.00055 and the protective channel at 0.9838 (maximum 0.9967 over the tested range). The multiplier types are reported with the crossings — a complex pair at the exact map's single 6.501 yr crossing (the Neimark–Sacker signature) and at the Euler extractive crossing (47.536 yr), and real -1 multipliers at the Euler crossings (79.143 yr extractive, 2.306 yr protective) — while the dominant multiplier's angle θ at the crossing and the continuous eigenvalue λ to which Section 3.2's transfer relation refers are not listed here; they are in the computational record (Appendix A). At a margin of this size the verdict is conditional on the monodromy's numerical construction — the stated precision record is the 200,001-point scan with bisection refinement; the linearisation construction and floating-point detail are archive items — and on the unlisted parameter vector (Table 3). What the margin does not affect is the ordering across updates and channels (exact 1.00035 below Euler 1.00055; protective 0.9838 below both) or the crossing directions; the annual-instability verdict is read at that status.

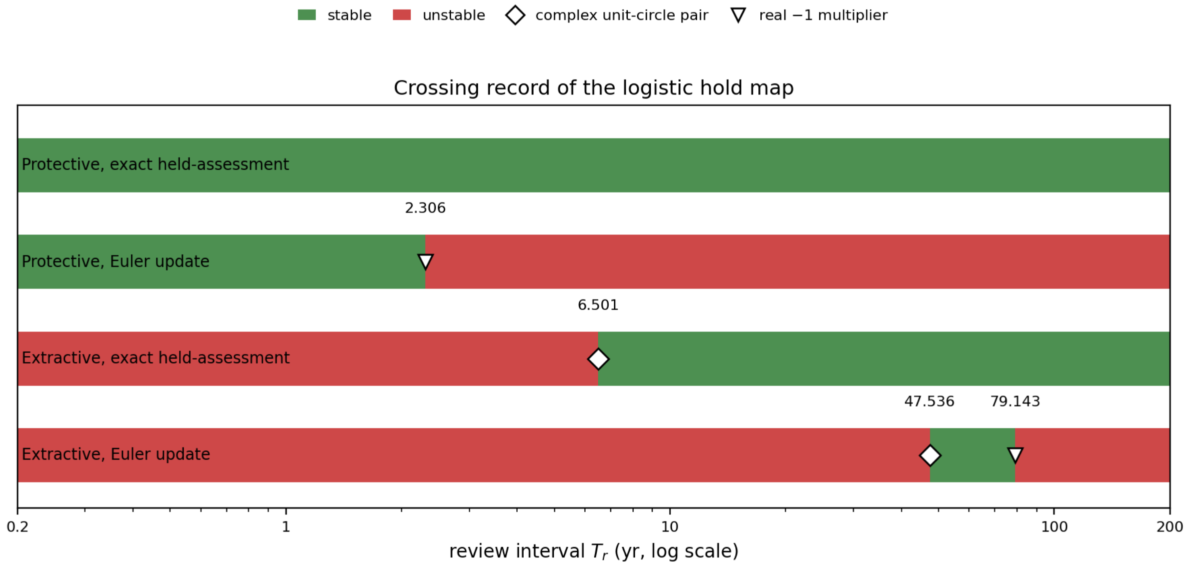

Figure 1: Crossing record of the logistic hold map: stable (green) and unstable (red) review-interval ranges for the four update × channel combinations (forward-Euler and exact updates on the extractive and protective channels), with crossing markers (diamond: complex unit-circle pair; triangle: real −1 multiplier). Annual review is nominally unstable on both extractive updates ($\rho = 1.00035$ exact, 1.00055 Euler; protective 0.9838), conditional on the numerical construction and parameter vector (Spectral margins paragraph); the Euler-reported 47.536 and 79.143 yr crossings are command-step artefacts relative to the exact update's single 6.501 yr crossing, and the protective Euler crossing at 2.306 yr is likewise an artefact — the exact protective map is stable at every tested interval (maximum spectral radius 0.9967).

The one-plant operator contrast. With the crossing record complete, the operator comparison can be made on one plant rather than across plants. The same extractive loop under continuous delay carries the companion delay study's certified Hopf pair at 3.666 and 150.358 yr (Abaee, 2026); the sampled operator's exact crossing sits at 6.501 yr. The companion's window is delay-stabilising, not delay-destabilising: the undelayed loop is already unstable, the lower crossing at $\tau_- = 3.666$ yr is stabilising, the upper crossing at $\tau_+ = 150.358$ yr destabilising, and the loop is linearly unstable for $0 < \tau < \tau_-$ (Abaee, 2026). Annual review is nominally unstable under sampling ($\rho = 1.00035$, conditional on the numerical construction and parameter vector as stated above), and the same loop under continuous delay is likewise unstable at $\tau = 1$ yr; at $T_r = 8$ yr both operators are stable (the sampled loop restabilised above its 6.501-yr crossing, the continuous loop inside its stabilised window). The operator contrast on one plant is an edge contrast: the sampled operator's restabilising crossing (6.501 yr) sits above the continuous operator's stabilising entry (3.666 yr), and the sampled loop's stability persists to the 200-yr scan ceiling where the continuous loop has already destabilised at τ_+ — the operator moves both edges of the stability window on the same plant, and the cross-plant observation of Section 2.3 is a same-plant one.

The undelayed limit, reconciled explicitly. Three statements of this manuscript bear on the undelayed loop and are read together. Section 2.3 states the continuous-delay equation and the logistic hold map to be the same feedback loop under two delay operators; Section 3.2 states that under hyperbolicity the sampled multipliers satisfy $\mu_j(T_r) = 1 + T_r\lambda_j + O(T_r^2)$, so that small-$T_r$ sampled stability follows the sign of $\operatorname{Re}\lambda_j$; and this section records that the undelayed equilibrium of the hold-map core is already unstable (annual $\rho = 1.00035$). The companion's undelayed record fixes the remaining sign: the undelayed linearisation of the same loop violates the Routh–Hurwitz condition, so $\operatorname{Re}\lambda > 0$ — the direction the sampled record suggests through Section 3.2's relation, with the sampled instability persisting down to $T_r = 0.2$ yr. Under that sign the records close at the zero-delay limit: the loop is unstable under both operators at zero delay and at $\tau = 1$ yr, and the stabilising crossing at 3.666 yr is the first crossing (there are no crossings in $(0, \tau_-)$; this even count is consistent with the stability-switch principle because the stability state does not change between zero delay and the first crossing) — no additional crossings are required. On this plant, the two operators differ only in where they place the window's edges, as the one-plant contrast above shows. The eigenvalue λ is not listed here (the spectral-margins record above; Appendix A); it is a derived eigenvalue, not a parameter, and its sign is read from the companion's undelayed record, and the operator-scoped records stand as stated — annual instability under sampling, instability under continuous delay at $\tau = 1$ yr, and the same-plant edge contrast — with Section 2.3's same-loop statement and Section 3.2's transfer statement consistent with both.

The protective controller on the same maps. The protective comparator run (sign-reversed response; linearised gains $C_E = -0.850336$, $C_Z = -1.661702$) on the same plant gives exact-update spectral radius 0.9838 at annual review and stability at every $T_r \in [0.2, 200]$ yr (maximum 0.9967). The Euler-run protective instability is an artefact band, not a property of the protective architecture.

Executed comparator evaluation at model level. The protective and fixed-plan comparators are run through the closed loop on the logistic hold-map core (nonlinear plant, $N_0 = 0.95\,N^*$, 800 reviews, metrics on the last 60%, multiplicative assessment error $\{0, 0.3\}$, seeds fixed; the spectral record remains the authoritative result, and the prospective real-system design of Section 4.5 remains unexecuted). Under 30% assessment error with ten seeds per cell: the extractive Euler core crashes to extinction in all ten seeds at $T_r = 8$, 12, and 20 yr, and at annual review its mean depletion frequency (fraction of post-transient reviews with $N < N^*/2$) is 0.22 with mean minimum stock $0.22\,N^*$; the extractive exact update never crashes, with mean depletion falling from 0.07 ($T_r = 1$) to 0.001 ($T_r = 20$); the protective exact update holds the equilibrium at every interval (depletion 0, mean minimum $0.89\,N^*$, rms stock deviation $\leq 0.001K$); the protective Euler update crashes in all ten seeds at $T_r = 20$ yr and shows depletion 0.9996 at $T_r = 12$ — the artefact operating at the trajectory level; in the deterministic baseline with fixed effort set at the equilibrium value, the fixed plan remains at equilibrium — a baseline rest point, not a robust performance result. The linear trajectory distortion between the Euler and exact updates grows with T_r (scaled-norm RMSD ≈ 2 at $T_r = 0.5$ yr, 10^4–10^5 at $T_r = 5$–40 yr, with the Euler path diverging in its artefact bands): the command step is not a small distortion. On the archived stage-structured review map, annual review is stable at every tested response value, and the archived record places the anchovy-class window at $T_r \approx 3$–4 yr and the sprat-class window at $T_r \approx 6$–12 yr, reporting robustness to 30% multiplicative assessment error. These are archived, unreproduced statements (Section 3.3): the generating computation is not available, and the pre-registered reconstruction below does not reproduce the multi-year windows; they are claims about the archived map, not results of the reconstructed object, and neither transfers to the other operator.

Two consequences follow. First, in the reported simulations relative effort excursions were larger than relative biomass excursions, and any quota-utilisation reading of the signal would require a quota, realised catch, and compliance model, which are not part of the model core. Second, no qualifying example was found in the case search of Section 2.6 of a real small-pelagic system operating a responsive 3–4 yr review, so the window prediction is untested, not falsified. The corresponding qualitative test for slow-regenerating resources is whether a responsive multi-year plan, adjusted against measured change, produces large-amplitude extraction cycles that a frozen multi-year cap does not (Section 4.5).

The stage-map reconstruction record. The stage-output values carry a stated limitation: the stage map's computational record — solver configuration, initial histories, stage registration — is not attached, so those values are not reproducible numerical propositions (Section 2.2). To supply the stage operator's boundary record, a new stage map was specified and pre-registered (plan dated and fixed 2026-09-01, before any run; parameter values and comparison criteria fixed in the plan), and its complete multiplier and trajectory records are reported here. The archived stage-output records are trajectory-classified only (Section 2.2). The reconstruction is a newly specified object: it is not claimed to be the object that produced the archived windows, and the comparison below is a post-hoc consistency check, not a validation of those windows.

Plant. A two-stage delayed-recruitment map with annual steps, $A_{t+1} = s_A A_t + s_J J_t - qEA_t$, $J_{t+1} = f(A_{t-\tau})$ with Beverton–Holt recruitment $f(A) = \alpha A/(1+\beta A)$, $s_A = s_J = e^{-M}$, and the pair (α, β) fixed by steepness and scale: $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$, $A_0 = 100$ — a scale convention matching the logistic core's carrying capacity. Surplus production $S(A) = s_J f(A) - (1-s_A)A$ replaces the logistic surplus in the deficit signal of equation (2). Every parameter is a cited literature value or a stated convention; none was chosen with reference to the archived windows. The class parameter sets: anchovy $M = 0.90$ yr^{-1} and recruitment delay $\tau = 1$ yr (Pauly and Tsukayama, 1987); sprat $M = 0.40$ yr^{-1} (mid-range of the Baltic SMS keyrun values, 0.3–0.8 yr^{-1}; ICES, 2020) and $\tau = 2$ yr (age at 50% maturity in the Baltic sprat assessment); cod $M = 0.20$ yr^{-1} (the fixed cod-assessment convention; ICES, 2021) and $\tau = 5$ yr (northern cod age at 50% maturity; DFO, 2011); slow-stock $M = 0.045$ yr^{-1} and $\tau = 25$ yr (orange roughy; Branch, 2001). Steepness $h = 0.75$ throughout — the standard default convention when no stock-specific estimate is available — with a sensitivity layer $h \in \{0.6, 0.9\}$. The controller of Section 2.1 is unchanged: equations (3)–(4) in discrete form with T_r varied over $\{1, \ldots, 50\}$ yr at annual internal steps, softplus sharpness $k = 10$ (the comparator machinery's value), contemporaneous exact assessment, and catchability $q = 0.001$ (the hold-map core's value) with a sensitivity case $q = 0.1$ (the archived stage record states no q; both values are reported for the reconstruction). The derivative construction is central finite differences of the review map at the fixed point (step 10^{-6}, chain-rule cross-check to 10^{-4}), and the scan grid is $T_r \in \{1, \ldots, 50\}$ yr at annual internal steps — finite-grid by construction, as the archived record itself was. The extractive channel is primary; the protective channel is computed with quota-tracking gains of Section 2.1 in the linearised review map, the same convention as the logistic protective record.

Records. At the specified controller value every class is stable at annual review — spectral radii $\rho(1) = 0.895$ (anchovy), 0.923 (sprat), 0.956 (cod), 0.994 (slow-stock) — reproducing the archived stage map's annual stability at the tested response value. The protective channel is stable everywhere on the grid (maximum spectral radius 0.968), mirroring the logistic protective record. The extractive channel's only instability at the specified value is a long-horizon band from $T_r = 34$–35 yr to the grid's end for the three faster classes (maximum spectral radius ≈ 1.98 at $T_r = 50$), with no crossings anywhere for the slow-stock class ($\rho(50) = 0.67$); the band is insensitive to the steepness layer. The trajectory classification (2000 review-steps from the specified initial condition — plant at the unfished equilibrium, memory filled, effort at half the equilibrium — tail of 500, relative tail standard-deviation thresholds 2% and 0.1%) agrees with the spectral record for the three faster classes, which converge at every $T_r \in [1, 20]$. In the multiplier-unstable long-horizon band ($T_r \geq 34$ yr) the faster-class trajectories remain biomass-converged or weak (relative tail standard deviation at most 0.6%) while effort excursions grow from approximately 7–15% at band entry to approximately 190% at $T_r = 50$: the instability manifests in the effort channel, to which the classifier — a biomass-tail threshold — is insensitive. The slow-stock class does not agree with its own multiplier record: it shows a persistent oscillation at $T_r = 10$ yr only (relative tail standard deviation 5.5%; dominant period ≈ 24 yr in both stock and effort; effort excursion 537% against a biomass excursion of 18%), converging at every other grid point, against a multiplier record with no crossings anywhere on the grid. A persistent oscillation at a linearly stable fixed point is bistability, a non-decayed transient, or a classification-threshold artefact; the record does not distinguish among them, and the cell is reported as a disagreement between the reconstruction's two records, not as an agreement, and is not used inferentially. Under 30% multiplicative assessment error the anchovy and sprat cells remain converged (maximum relative tail standard deviation 0.0009); the cod cells weaken at $T_r = 10$ and 20 yr (converged $\rightarrow$ weak, maximum 0.0025) while its annual-review cell stays converged (0.0004); and the slow-stock class oscillates persistently at the long-horizon cells $T_r = 30$–50 yr (2.1–2.6%, against a stable multiplier record without error — cells read as noise-driven variance rather than oscillation, since the 2% relative tail threshold is not noise-adjusted) while its $T_r = 10$ cell weakens from its error-free oscillation and its $T_r = 20$ cell weakens (0.017). The catchability sensitivity analysis is informative: at $q = 0.1$ every class is unstable at annual review ($\rho(1) \geq 1.29$); the three faster classes destabilise on $[1, 18-31]$ yr, restabilise, and re-enter an unstable band at 36–42 yr, while the slow-stock class is unstable across the entire grid ($\rho(50) = 7.8$) — the reconstructed stage loop's stability profile depends strongly on the catchability scale, which is unspecified for this plant.

Comparison with the archived windows (criteria pre-registered; verdicts from the first complete run):

Criterion (archive)	Reconstruction	Verdict
anchovy persistent oscillation at $T_r = 3$–4 yr	converged at both	MISMATCH
anchovy weak response at $T_r = 2$ yr	converged	MISMATCH
anchovy annual convergence	$\rho(1) = 0.895$	MATCH
sprat persistent oscillation at $T_r = 6$–12 yr	converged throughout	MISMATCH
cod convergence for every $T_r \in [1, 20]$	converged throughout	MATCH
slow-stock oscillation below 20–30 yr, convergence at 30–50 yr	oscillation at $T_r = 10$, convergence at 30–50	MATCH

Reading. The reconstruction reproduces the archived record's qualitative features — annual stability on the stage map, the protective channel's stability, cod-class convergence over the 1–20 yr grid, the slow-stock oscillation-then-convergence pattern, and the effort-versus-biomass excursion ordering — but not the anchovy 3–4 yr or the sprat 6–12 yr response regions: on this specified plant family those classes converge at every review interval, and the only instability the reconstructed loop produces is its own long-horizon band, entering at 34–35 yr and extending to the grid's end, which has no counterpart among the archived windows. The comparison, however, is uninformative rather than a non-reproduction. The stage map states no catchability; the table is computed at the imported value $q = 0.001$; and at the sensitivity case $q = 0.1$ every verdict in the table flips — every class unstable at annual review ($\rho(1) \geq 1.29$), the slow-stock class unstable across the entire grid — so a match or a mismatch at either value is a statement about this specified family at an unspecified scale, not an adjudication of the archived values, whose generating computation is not available. The reconstruction therefore supplies the stage operator's boundary record as a new, fully specified object, whose comparison with the archived windows is a consistency check rather than a validation.

3.5 The selected 42-stock spectral screen

No stock in the screened cohort has a peak in the specified biomass or effort band meeting all robustness criteria. Baltic sprat, for example, has a biomass coefficient of variation of 0.40, but its variation is low-frequency and regime-dominated rather than a robust target-band peak. The null carries a three-way restriction: it is not proof of absence, not a comparison of controller signs, and not causal evidence that annual review stabilises anything. On the stage-structured review map, annual-review stability at every tested response value is consistent with this null; consistency is not a test. The institutional reading is therefore limited: under this screen, target-band cycles are not common in assessed-stock records, and any claim that periodicity itself diagnoses an institutional feedback is unsupported by this cohort.

3.6 Power of the injected-signal screen

On 100–200 yr synthetic records, the sprat-class signal has estimated power 1.0 at noise scale $\sigma = 0.1$ and approximately 0.24–0.58 at $\sigma = 0.3$. The anchovy-class effort signal has power approximately 0.02–0.14 over the tested horizons and noise levels; its longer effort peak and slow amplitude growth can offset its larger relative excursion. These are conditional simulations: no minimum-power guarantee holds per empirical stock, and the 100–200 yr horizons exceed many eligible series. The anchovy-class null is consequently weakly informative under this test, and the sprat-class result is informative only in favourable noise and record-length regimes. The empirical screen is therefore a selected-cohort consistency check and the baseline for a prospective design, not a general test of the extractive mechanism.

The evidentiary separation is complete: the stage-dependent regions, noise experiments, and power estimates do not establish population-wide frequencies or policy effects and cannot be transferred to the extractive hold-map controller or to protective control. A complementary prospective design estimates resolvable gain and phase rather than waiting for several complete endogenous cycles. For a locally linear specified model, the empirical target is the frequency response from an independently timed assessment or command perturbation to realised effort and stock response over the annual-to-decadal band the data support. Identification requires exogenous excitation, an intervention design, or a justified closed-loop method (Forssell and Ljung, 1999). The analysis reports phase margin and uncertainty, compares alternative ecological and controller factorisations, and rejects the mechanism when no admissible factorisation reproduces the pre-registered gain–phase curve. For the anchoveta–ENSO association of Section 3.7, the planned confirmatory tests are: the era-split attribution — explaining the post-1985 attenuation of the ENSO–catch coupling (management change, reporting regime, or biological reorganisation) — as the focal test; a mechanistic pathway model from the SOI through coastal temperature and upwelling to recruitment; and the index–lag specification (SOI, lags zero to two, both stocks), fixed by the executed battery.

3.7 The zero-count case search

No candidate satisfied all four criteria after primary-source and station-level review. Bangkok (durably responsive institutional feedback: the groundwater-extraction control regime persisted across multiple review cycles rather than being a one-time cap or ban) and La Mancha Oriental (on the stabilising side before its 2019–2023 extraction relapse) are the closest cases. The unconfounded delay oscillators identified in produced-capital systems (livestock and electricity-capacity cycles) do not contain the autonomously regenerating stock the resource model assumes. The per-system calculations use archived input series and are reported at that status in the Supplementary material (S4); three groups illustrate the screening logic.

Sheridan-6 groundwater shows a decline–recovery–decline pattern with precipitation explaining approximately half the index-well variance ($R^2 \approx 0.47$). The official programme record establishes a 55 acre-inch block allocation over the five-year first period rather than a continuously adjusted feedback. Icelandic cod under the 1995 harvest-control rule (annual total allowable catch at 25% of fishable biomass, subject to a minimum catch provision) has a post-rule coefficient of variation of 0.387 (calculated here) with a 10–15 yr fluctuation. The estimated implementation lag, however, is approximately 0.2–0.3 yr — a lag that is not a review interval — and cohort resonance supplies an alternative mechanism, its period 15–25 times shorter than the four-state prediction — the stage-structured review map's closed loop of four state components (adults A, juveniles J, memory signal Z, held effort E), whose slow-stock class carries the centuries-scale dominant timescales of Section 3.3. Icelandic haddock under a related rule has a post-implementation coefficient of variation of 0.143, lower despite higher recruitment variability.

Peruvian anchoveta provides a further discriminator. The 1950–2019 catch series has a robust period near 3.7 yr, matching ENSO recurrence as a candidate driver, with cross-correlation $|r| \approx 0.31$ for ENSO leading catch. A confirmatory battery on the archived Sea Around Us series behind that figure (Peru and Chile, 1950–2019; Supplementary S4) reproduces the 3.7-yr peak (3.70 yr, co-dominant with 7.96 yr in a flat multi-peak spectrum) and resolves the association to the Southern Oscillation Index: Peru–SOI $r = +0.51$ contemporaneous ($p < 0.0001$) and +0.42/+0.40 at one- and two-year lags, with the Chilean series replicating ($r = +0.39$ at lag two, $p = 0.001$). All ten of the ninety tested index–lag cells that survive the Benjamini–Hochberg bound are SOI cells, so the SOI association is multiplicity-robust where no other index is. The ninety index–lag cells define the Benjamini–Hochberg multiplicity family; the Granger and split-half tests are confirmatory and outside that family. Bivariate Granger tests find one-sided ENSO-to-catch dependence in both stocks (Peru $p = 0.00009$, Chile $p = 0.00016$, at lag two; reverse directions $p \geq 0.19$). The association is not era-invariant: split-half analysis confines it to 1950–1984 (lag-one $r = +0.42$, $p = 0.013$; after 1985 it is absent, $r = +0.13$, n.s., with the lag-two cell reversing sign), and the early-period strength survives both the exclusion of the three collapse years and a reported-catches-only sensitivity on the Chilean series, so the attenuation is not a reconstruction artefact. Convergent cross mapping remains directionally inconclusive even at seventy annual points, and the post-1985 attenuation is treated as a focal test rather than claimed as a regime effect. The association is therefore evidence of shared, era-bounded periodicity, not an identified mechanism — its mechanistic-scale support is the documented El Niño–anchoveta pathway (Chávez et al., 2003) and the sediment fish-scale records (Gutiérrez et al., 2009) — and the subannual review regime lies below the anchovy-class review-interval response region, so the unclassified controller prevents that comparison from testing the mechanism.

Two structural hypotheses may help explain the zero count. The long records needed to test the claim exist primarily for visible crises, and visibility is correlated with a fast non-institutional driver. And on the continuous-delay parameterisation that produces the instability window ($rg \approx 1.5$–1.6) the predicted period is centuries — longer than any institution has held a fixed response rule — while the 3–12 yr windows belong to the stage-structured map. The two period scales are properties of different operators and should not be read as one prediction. The visible-record pattern — long series repeatedly associated with climate variability, cohort effects, infrastructure changes, or emergency interventions — suggests, without establishing, a selection mechanism in which systems receive intensive monitoring after complex crises. The retrospective search cannot distinguish that hypothesis from the null of coincidental association, and no causal identification of selection bias is claimed.

One documented system sits at the window's edge rather than inside it. The spruce budworm's 30–40 yr outbreak cycle maps to an effective regeneration rate $r \approx 0.025$–0.033 yr^{-1} under a 1/period reading (Ludwig, Jones, and Holling, 1978) — above the upper edge of the companion delay study's baseline instability window (≈ 0.022 yr^{-1} at $\eta = 0.914$; Abaee, 2026) and inside it only at elevated effort response, whose window extends to ≈ 0.061 yr^{-1} at $\eta = 3$. The cycle is endogenous predator–prey outbreak dynamics rather than institutionally driven: the system corroborates the mechanism class, not the institutional claim.

3.8 The northern cod two-window split

The case's positive content is a descriptive partition. **The data split the phenomenon into two events — the crash interpretation is formulation-dependent, and the post-collapse dynamics expose a second identification problem not resolved by the**

mortality-allocation comparison. The split is the positive content, not a new mechanism.

The crash window (1991–1995) is documented by the assessment table (DFO CSAS SAR 2016/026, Table A2; Table 2): spawning stock biomass falls from 735 to 10 kt across the five calendar years 1991–1995 while estimated natural mortality reaches 2.2–2.6 yr^{-1}, roughly ten times pre-collapse levels (the pre-collapse reference period, estimator, and uncertainty interval for this ratio are documented with the assessment table; Table 2 begins in 1991). The displayed values are rounded values from the assessment table (e.g. underlying values 381.95, 101.05, 30.55 kt), and the survival column $\exp(-M)$ is a transformation of the reported instantaneous mortality estimate, not an independently observed survival series. The interpretation of the crash is a formulation attribution, not a causal claim. The NCAM M-shift formulation allocates most estimated mortality to natural death, and NCAM's M is an estimated unobserved-death component conditional on model structure — assessment-framework proceedings explicitly caution that unreported fishing deaths may enter de-facto M (Cadigan, 2016). The constrained-M formulation attributes the crash to unreported catch. The constrained-M quantities (crash window $M = 0.46$, $F = 1.37$, unreported catch 257.8 kt yr^{-1} — 1.025 yr^{-1} relative to mean spawning biomass (a yearly flow divided by a stock); non-recovery window $M = 0.43$, $F = 0.25$, 3.7 kt yr^{-1}) are unreproduced: they are targets for future reproduction requiring equations, source series, code, units, windows, and uncertainty, listed as open problems and stated here as hypotheses, not results.

Table 2. Northern cod spawning stock biomass (SSB) and estimated instantaneous natural mortality (M) from DFO CSAS SAR 2016/026 (Table A2, NCAM M-shift formulation), rounded from the assessment table. The survival column $\exp(-M)$ is a transformation of the reported M — annual survival conditional on the estimated natural-mortality component alone, since fishing mortality and other modelled removals also apply — not an independently observed survival series.

Year	SSB (kt)	M (yr^{-1})	$\exp(-M)$
1991	735	1.002	0.367
1992	382	2.214	0.109
1993	101	2.575	0.076
1994	31	2.331	0.097
1995	10	0.288	0.750
1996	16.05	0.341	0.711
2000	34.42	0.717	0.488
2004	20.07	0.362	0.696
2005	25.18	0.288	0.750
2010	96.91	0.696	0.499
2015	298.65	0.278	0.757

The non-recovery window is the second event. After the moratorium, the series both rises and falls across tens of thousands of tonnes — 16.05, 34.42, and 20.07 kt in 1996, 2000, and 2004, before the recovery window of 25.18, 96.91, and 298.65 kt in 2005, 2010, and 2015 (selected years; the full annual record with uncertainty intervals is the assessment table). The phase-line obstruction of Section 2.7 applies as a model-class diagnostic (not an empirical rejection of scalar autonomous models for northern cod): an error-free trajectory with repeated direction reversals cannot arise from a fixed scalar autonomous model with constant removals. The assessment record does not satisfy the diagnostic's two conditions — the SSB series is an assessment estimate rather than an exact trajectory, and removals were not fixed through the window — so the proposition is a statement about model adequacy for this class, not a direct empirical rejection. Two scope statements govern the result. The reliable contradiction is repeated direction reversal: convergence toward K can be arbitrarily slow near degeneracy, so failure to reach an unspecified biomass by an unspecified deadline is not a contradiction. And the obstruction must be separated from rejection under measurement error, process noise, age structure, migration, time-varying mortality, and state-space observation models — the incompatibility is a property of the exact trajectory class, nothing broader.

The subsequent record does not close the second window. Rose (2026), comparing two surplus-production reconstructions spanning 1983–2023 (Rose and Walters, 2019; DFO, 2024), documents that surplus production and stock growth stalled after 2015, with some years negative, and that the stall-point biomass remains controversial. Structural-equation analyses in the same review attribute the production deficit to capelin limitation and harp seal predation rather than fishing alone. For the present analysis the key point is the persistence of the non-recovery: a decade after the rebound reported by Rose and Rowe (2015), the post-2015 production stall is inconsistent with the simplest fixed-parameter surplus-production interpretation considered here, which is precisely what the split's second window — a second identification problem not resolved by the mortality-allocation comparison — states.

The ecosystem context is background only: between 1985–87 and 2013–15, harp seal biomass rose from 49,600 t to 161,183 t (a 3.2-fold increase) and capelin biomass fell from 13.77 to 4.97 t km^{-2} (a 64% decline) in the mass-balance record (Tam and Bundy, 2019). These are descriptive mass-balance inputs, not causal tests.

4 Discussion

4.1 Architecture substitution: what changes when the operator changes

The operator-spectra results of Section 3.4 are the paper's core methodological finding. The same feedback loop — surplus production, an institutional deficit signal, an effort law — carries an archived, unreproduced instability record near 3–4 yr of review under the stage-structured map (a different ecological plant, Section 2.3; the archived windows' provisional status and the reconstruction's comparison with them are stated in Sections 3.3 and 3.4; the plant–operator confound is stated alongside — the operator effect is not claimed to be isolated by this comparison), convergence over a 1–20 yr grid under the cod-class parameterisation of that same map, and instability at annual review with a complex unit-circle crossing near 6.5 yr under the exact update — the Euler-reported crossing near 47.5 yr being a command-step artefact relative to the exact update — under the logistic hold map. The stage-map layer of that record has its own stated limitation: the stage map fixes no catchability, the reconstruction imports the hold-map core's $q = 0.001$, and at the sensitivity case $q = 0.1$ every verdict of the Section 3.4 comparison flips — every class unstable at annual review ($\rho(1) \geq 1.29$), the slow-stock class unstable across the entire grid ($\rho(50) = 7.8$) — so the archived-window comparison is uninformative at the unspecified scale rather than a non-reproduction (Section 3.4's Reading).

The complete crossing record sharpens the operator finding: On one plant the exact update has exactly one crossing (extractive, 6.501 yr) and none (protective), while the Euler update reports artefact bands. The review cadence is a local spectral design parameter, and the command step is a second, independent one. None of these statements can be obtained from the continuous-delay equation, and none transfers to it. Any empirical or theoretical claim about institutional feedback must therefore state its operator: a stability window located under continuous delay is not a prediction about an institution that reviews periodically, and an annual-review result does not generalise to multi-year plans. The governance architecture itself is a testable component of the hypothesis, not a re-parameterisation detail.

4.2 What the null does and does not establish

The 42-stock screen returns a spectral null, and its value lies in what it refutes: the claim, implicit in treating periodicity as evidence, that robust target-band cycles are common in assessed-stock records. It does not establish the contrary. With anchovy-class power between 0.02 and 0.14 across tested noise and record-length regimes, the screen cannot adjudicate the extractive mechanism for that class; the sprat-class result is informative only in favourable regimes; and no screen of retrospective series can compare controller signs, because the sign is not in the data. This is the same null discipline that the skeptical re-analysis of the Russell Cycle applied to climate-forced cycles (McManus, Licandro, and Coombs, 2016), transferred to governance-feedback claims: a diagnostic is not a causal claim, and no accumulation of diagnostics converts to causal evidence.

4.3 The cod case at its exact status

The cod case contributes a descriptive partition, not a mechanism. The crash-window mortality is formulation-dependent because the NCAM M-shift allocation is that — an allocation of unobserved deaths conditional on model structure (Cadigan, 2016). The post-collapse record is not resolved by either mortality-allocation formulation because no single fixed-regime model reproduces the repeated reversals of the post-moratorium record (the phase-line obstruction, applied as a model-class diagnostic under Section 3.8's two conditions), and the post-2015 production stall documented by Rose (2026) shows the second window persisting under independent reconstructions. The obstruction mathematics is likewise bounded: it concerns exact trajectories of the fixed-parameter, fixed-removals autonomous class and is not a rejection under measurement error, process noise, age structure, migration, time-varying mortality, or state-space observation models. What the case supplies is a falsification benchmark: a well-documented collapse-and-stall record against which prospective designs can be powered and scored.

4.4 A falsification standard for institutional-feedback claims

The model family gains empirical content by specifying the outcomes that count against its mechanism and parameterisation. Five outcomes:

1. If independently measured implementation and deployment lags do not covary with the instability bands predicted under fitted resource parameters, the quantitative delay claim is weakened.
2. If an observed oscillation remains after climate, cohort, and regime forcing are modelled, but its phase relation between decline signal, effort, and biomass contradicts the controller's causal ordering, the mechanism is rejected for that case.
3. If sampled-data analysis removes a continuous-delay band under realistic review rules, the continuous model cannot be used for that institution.
4. If a protective controller dominates the extractive controller across uncertainty without inducing the claimed volatility, no anti-regulation conclusion may be drawn.
5. If realistic record lengths have low power for the predicted signal, field spectral nulls cannot adjudicate the mechanism; prospective or experimental evidence is required.

4.5 Prospective designs

The retrospective evidence does not identify the institutional mechanism; the constructive content of this paper is the programme that could. Five designs are specified as preregistration targets; none has been executed, and each is intended for preregistration before outcome inspection — no registration identifier or archived protocol exists yet, and none is claimed.

Governance-event panels. For each resource–jurisdiction unit, construct a source-linked event record containing the raw-observation date, the public or scientific recognition date, assessment completion, scheduled review, formal decision, legal adoption, physical deployment, compliance, realised-pressure change, and subsequent ecological-response dates. Date uncertainty, interval censoring, revisions, missing stages, and overlapping interventions are kept distinct rather than collapsed to one lag. Such a panel estimates component-specific delay distributions only for stages actually observed; otherwise the estimand is a combined interval or a closed-loop phase, not separate τ_{dec} and τ_{dep} values. The cod case supplies two dated decision events for such a panel — the moratorium announcement of 2 July 1992 and the reopening of 26 June 2024 with an 18 kt total allowable catch — with two negative constraints: no governance lead is inferred from annual biomass data, and a fast response does not establish adequacy.

Quasi-experimental timing. Candidate interventions include staggered adoption, discontinuities in review schedules, rule changes, jurisdictional borders, phased quota systems, and administrative reforms whose timing is plausibly independent of the outcome innovation. An event study, difference-in-differences design, synthetic control, interrupted time series, or state-space intervention model is chosen according to its assumptions, not merely data availability. Two coding rules are binding: a change in implementation lag is not coded as a change in T_r, and a nominal schedule change is not a treatment unless it changes a responsive decision opportunity.

Out-of-sample mechanism comparison and the displacement discipline. For each candidate system, compare at least an environmental-forcing model, a cohort- or demographic-resonance model, an institutional-delay model with specified controller sign, a combined model, and a null time-series model. Models make predictions before the held-out block is scored, using stated predictive and calibration criteria; model weights or posterior probabilities require an explicit likelihood and prior, and predictive ranking alone does not identify a causal mechanism. The displacement rule: a delay explanation is weakened when it cannot reproduce the preregistered phase ordering, and it is displaced when a competing mechanism predicts the held-out observations better — complexity is admitted only on scored evidence, never by accumulation.

Controlled and randomized human-in-the-loop experiments. A minimum design places participants in the same simulated renewable-resource environment and randomises review cadence and the timing or sign of decision feedback, with ecological shocks held common across arms where appropriate. Primary endpoints include realised pressure, threshold crossings, recovery time, variability, and the gain–phase relation between assessments, commands, and actions — the gain–phase signature being the more diagnostic empirical target than periodicity alone, since a spectral peak cannot identify the loop direction. Treatment rules, stopping and safety criteria, sample size, exclusions, and analysis are intended for preregistration; field pilots that would randomise extractive response to decline require viability constraints and stopping rules before controller-sign randomisation is considered, and the randomisation described is a laboratory or advisory-interface design, not a field intervention on a live stock. Agent-based institutional experiments, digital-twin exercises, and carefully governed field pilots complement this design; extrapolation from a laboratory or simulated resource to a field institution remains a separate external-validity claim. Mechanism-class precedent exists — controlled population and resource-management experiments show that delay- and parameter-driven transitions can be empirically studied (Costantino, Cushing, Dennis, and Desharnais, 1995), and commons-free fishery-management experiments in which subjects overshoot by approximately 60% from stock-and-flow misperception show the behavioural substrate (Moxnes, 1998) — but precedent for the mechanism class is not validation of the institutional equations.

Closed-loop management strategy evaluation. Because retrospective evidence does not identify the mechanism, policy comparison must be conducted in a closed loop that keeps process, observation and assessment, parameter, structural-model, decision, and implementation uncertainty distinct (the management-procedure tradition: Punt and Donovan, 2007). Each simulation replicate contains: an operating model (age- or stage-structured or other resource dynamics, environmental forcing, density dependence, structural alternatives); an observation model (survey and catch observations, missingness, bias, autocorrelated error); an assessment model (estimator, update frequency, retrospective bias, uncertainty); a decision rule (extractive, protective, fixed-plan, or hybrid, including caps on change and emergency clauses); an implementation model (compliance, deployment lag, effort creep, realised versus commanded pressure); and performance metrics (threshold risk, yield and service delivery, variability, closure frequency, effort cost, recovery time, distributional impacts). The core experimental design crosses

$$T_r \times \tau_{\text{dec}} \times \tau_{\text{dep}} \times \text{controller sign} \times \text{observation/assessment error} \times \text{parameter draw} \times \text{operating-model class} \times \text{process}$$

At minimum, responsive extractive, responsive protective, and fixed-plan controllers are compared; a conclusion that compares only two review intervals inside the extractive class cannot be generalised to governance as a whole. Parameter uncertainty varies quantities within a specified operating model; process uncertainty governs stochastic state evolution; observation and assessment uncertainty govern the information supplied to the rule; and structural uncertainty varies the operating-model class itself, represented by multiple operating models rather than one parameter covariance matrix around a fitted model. Model-class worst-case, distributionally robust, and model-averaged performance answer different questions and are reported separately.

4.6 Distributive constraints where reproducible

The social side of the cod case is presented at measurement level (Appendix B): the constituency-definition mismatch table (licence holders against census-subdivision residents; fishing income against all-resident sector income; the unspecified floor), the Statistics Canada community-level series (tables 38-10-0167-01 and 38-10-0168-01), the measured-floor construction (I_k, c_k) with its componentwise non-decline rule, and the admitted world-hook instruments are given in Appendix B, with the full instrument detail and the unreproduced pipeline register in the Supplementary material (S6). What the case does not measure is the point: no licence-holder panel exists, the community series is not one, and no measured floor is operational — the distributive constraints remain stated rather than resolved (Section 4.7 (viii)).

4.7 Limitations

(i) Diagnostics are not causal claims: the spectral null, the response regions, the power values, the case calculations, and the cod split carry their stated types and no more.

(ii) The stage-structured response regions are exploratory finite-grid, trajectory-classified records pending complete stage registration and multiplier analysis; the logistic hold-map multiplier record is complete (Section 3.4), and the stage operator's multiplier and trajectory record is supplied by the reconstruction of Section 3.4 — a newly specified object whose comparison with the archived windows is a consistency check, not a validation. No Poincaré-map multiplier classification is claimed for the archived stage-output values, and they are not reproducible numerical propositions until the original computational record is attached.

(iii) The two review-map operators are distinct: statements computed on the hold map, on the stage-structured map, and on the continuous-delay equation do not transfer to one another.

(iv) The screen is a selected-cohort consistency check whose power is high only in favourable noise and record-length regimes; the null is not proof of absence and adjudicates nothing about controller sign; the AR(1) null may understate low-frequency power.

(v) The zero-count case search does not independently disconfirm the mechanism: finding no eligible cases under these criteria is not evidence against it.

(vi) The cod case establishes a descriptive partition, and no mechanism is identified.

(vii) The obstruction mathematics concerns exact trajectories of the fixed-parameter, fixed-removals autonomous class and is not a rejection under measurement error, process noise, age structure, migration, time-varying mortality, or state-space observation models.

(viii) The social object is not operationalised: the human series is not assembled, the normative floors are unoperationalised, and documenting the gap is the finding.

(ix) The prospective designs are preregistration targets, not executed studies, and they do not retroactively change the status of the retrospective results.

5 Conclusion

Periodic review is the operating architecture of fisheries governance, and it is not well represented by either of its standard formalisations. The sample-and-hold model specified here makes the governance clock an explicit object: seven time objects, a projected explicit controller, and a spectrum computed per operator rather than per loop. Its empirical layer contains one case whose positive content is a descriptive split — the northern cod two-window partition — and a falsification discipline for the rest: a selected-cohort screen with no target-band discoveries and stated power, a zero-count case search with its structural hypotheses, and a falsification standard with five outcomes that count against the mechanism. The mechanism is falsifiable, and the five prospective designs specify how it would be falsified, comparatively supported, or left unresolved. Until those designs are executed, the summary of the empirical layer is the one stated here — a well-posed mechanism, an exploratory computational record, a selected-cohort screen with limited power, a zero-count case search that is not disconfirmation, and one case whose positive content is a descriptive split.

Appendix A. Registration and reproducibility record (consolidated)

This appendix collects the registration and reproducibility requirements stated in Sections 2.2, 2.4, 2.5, 3.3, and 3.4. The convention: an analysis is pre-registered when its plan is dated and fixed before any run, as the stage-map reconstruction of Section 3.4 is; computational artifacts (solver configuration, seeds, identifiers, calibration records) are archived with the materials. The consolidated requirements:

- The solver configuration and initial histories are registration requirements; until that computational record is complete, the stage-output values carry provisional status (Sections 2.2 and 3.3).
- The RAM stock identifiers and the eligibility table are registration requirements (Section 2.4).
- The full null-calibration record — AR(1) coefficient estimation, detrending inside each null replicate, missing-data treatment, and the number of Monte Carlo replicates — is archived with the computational materials (Section 2.4).
- The simulation code and seeds are registration requirements (Section 2.5).

Two further consolidated records. First, the members of the logistic-core parameter vector that the text does not list (r, K, $E_{\max}$, δ_0, Z_{ref}, Δ_{ref}, τ_m; $\delta = \log 2/10$ is stated in Section 2.1), the linearised fixed point (N^*, E^*, Z^*) and its interiority to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$, the monodromy's numerical construction, and the continuous eigenvalue λ and crossing angle θ of Section 3.4's records are part of the computational record; Table 3 collects the parameter values stated in the text. Second, the Data availability statement records the deposit status of the same requirements (including the reconstruction's dated plan, campaign code, and output tables); the prospective designs of Section 4.5 remain preregistration targets with no registration identifier or archived protocol.

Table 3. Parameter values stated in this manuscript for the logistic hold-map core and the stage-structured reconstruction. No value here is newly computed; values not stated in the text remain in the computational record (this appendix).

Object	Parameter	Value as stated	Stated at
Logistic hold-map core	catchability q	0.001	Section 3.4 (stage Plant paragraph, as the hold-map core's value)
Logistic hold-map core	softplus sharpness k	10	Section 3.4 (Plant paragraph)
Logistic hold-map core	effort-response coefficient η	0.914 (continuous-delay asides on the same loop)	Sections 3.3 and 3.7
Logistic hold-map core	r, K, $E_{\max}$, δ_0, Z_{ref}, Δ_{ref}, τ_m	not listed here	computational record (this appendix)
Logistic hold-map core	memory shift δ	$\log 2/10 \approx 0.069$ (equilibrium memory level)	Section 2.1
Logistic hold-map core	linearised fixed point (N^*, E^*, Z^*); interiority to the nonsmooth regions of Φ_k and Π	not listed here	computational record (this appendix)
Logistic hold-map core	crossing record	47.536, 79.143, 2.306, 6.501 yr; $\rho = 1.00035$, 1.00055, 0.9838, 0.9967	Section 3.4
Logistic hold-map core	scan construction	200,001-point scan with bisection over $[0.2, 200]$ yr	Section 3.4
Stage reconstruction	class natural mortality M and recruitment delay τ	anchovy (0.90, 1), sprat (0.40, 2), cod (0.20, 5), slow-stock (0.045, 25) yr	Section 3.4 (Plant paragraph, with sources)
Stage reconstruction	steepness h	0.75; sensitivity layer $\{0.6, 0.9\}$	Section 3.4
Stage reconstruction	scale and survival	$A_0 = 100$; $s_A = s_J = e^{-M}$; $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$	Section 3.4
Stage reconstruction	catchability q	0.001; sensitivity 0.1	Section 3.4
Stage reconstruction	softplus sharpness k	10	Section 3.4
Stage reconstruction	derivative construction	central finite differences, step 10^{-6}, chain-rule cross-check to 10^{-4}	Section 3.4
Stage reconstruction	scan grid	$T_r \in \{1, \dots, 50\}$ yr at annual internal steps	Section 3.4
Stage reconstruction	trajectory classification	2000 review-steps, tail 500, relative tail standard-deviation thresholds 2% and 0.1%; initial condition at the unfished plant equilibrium, memory filled, effort at half the equilibrium	Section 3.4

Appendix B. Distributive constraints where reproducible

Where the social side of the cod case can be presented, it is presented at measurement level. The worst-off relevant population is a modelling choice — a named constituency — and the candidate constituency for 2J3KL is the registered inshore harvesters and licence holders. The measurement record against that candidate is the following mismatch table, each row stated rather than resolved:

Object	Candidate definition	Available measure	Mismatch	Required data
Population	registered licence holders	census-subdivision residents	not the same population	licence microdata
Income	fishing income	all-resident sector income (incl. aquaculture, processing)	includes non-fishing income	administrative income
Floor	specified cutoff (I_k, c_k)	none operational	not specified	specified instrument and cutoff

The available community-level series (Statistics Canada tables 38-10-0167-01 and 38-10-0168-01; mean income from fishing falling from 32.2% to 25.6% across 43 fishing-dependent census subdivisions, 2016–2021) is all-resident employment income including aquaculture and processing. The constituency must therefore either be redefined to the census-subdivision populations, or licence-holder microdata or administrative data must be obtained — the community table is not a licence-holder panel. A measured floor is a pair (I_k, c_k) — an instrument and a cutoff — a measurement object distinct from the norm that motivates it, and the componentwise rule is that the arrangement fails as soon as any measured margin $m_k(G,t) = I_k(G,t) - c_k$ is negative; the conjunction admits no compensatory master scalar. Non-decline is a normative rule, not automatically an empirically justified floor: baseline, cohort, inflation and purchasing-power treatment, uncertainty, attrition, acceptable variation, authority, and structural diversification all must be specified before a floor measures anything. The instruments admitted as world-hooks (the global Multidimensional Poverty Index of Alkire and Foster, 2011; the World Bank Poverty and Inequality Platform) carry release and vintage locks, and what they cannot capture — who pays for persistence, who may use remaining variety, whose knowledge counts, future persons — is stated as a boundary and left open rather than filled with a scalar score. The full instrument detail and the unreproduced pipeline register are in the Supplementary material.

References

Abaee, A. 2026. Delay-induced regime change in harvested stocks: the mobilising and protective channels of institutional feedback, and the review interval as control. Zenodo. https://doi.org/10.5281/zenodo.22554217.

Alkire, S., and Foster, J. 2011. Counting and multidimensional poverty measurement. Journal of Public Economics, 95: 476–487.

Ashwin, P., Wieczorek, S., Vitolo, R., and Cox, P. 2012. Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system. Philosophical Transactions of the Royal Society A, 370: 1166–1184.

Benjamini, Y., and Hochberg, Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society B, 57: 289–300.

Benjamini, Y., and Yekutieli, D. 2001. The control of the false discovery rate in multiple testing under dependency. Journal of the Royal Statistical Society B, 63: 425–441.

Branch, T. A. 2001. A review of orange roughy Hoplostethus atlanticus fisheries, estimation methods, biology and stock structure. South African Journal of Marine Science, 23: 181–203.

Cadigan, N. G. 2016. A state-space stock assessment model for northern cod, including under-reported catches and variable natural mortality rates. Canadian Journal of Fisheries and Aquatic Sciences, 73: 296–308.

Chávez, F. P., Ryan, J., Lluch-Cota, S. E., and Niquen, M. 2003. From anchovies to sardines and back: multidecadal change in the Pacific Ocean. Science, 299: 217–221.

Cohen, J. 1988. Statistical Power Analysis for the Behavioral Sciences, 2nd edn. Lawrence Erlbaum, Hillsdale, NJ.

Costantino, R. F., Cushing, J. M., Dennis, B., and Desharnais, R. A. 1995. Experimentally induced transitions in the dynamic behaviour of insect populations. Nature, 375: 227–230.

DFO. 2011. Recovery potential assessment for the Newfoundland and Labrador Designatable Unit (NAFO Divs. 2GHJ, 3KLNO) of Atlantic Cod. DFO Canadian Science Advisory Secretariat Science Advisory Report 2011/037.

DFO. 2016. Stock Assessment of Northern cod (NAFO Divs. 2J3KL) in 2016. DFO Canadian Science Advisory Secretariat Science Advisory Report 2016/026.

DFO. 2022. Stock Assessment of Northern cod (NAFO Divs. 2J3KL) in 2022. DFO Canadian Science Advisory Secretariat Science Advisory Report 2022/041.

DFO. 2024. NAFO Divisions 2J3KL Northern cod (Gadus morhua) stock assessment to 2024. DFO Canadian Science Advisory Secretariat Science Advisory Report.

Forssell, U., and Ljung, L. 1999. Closed-loop identification revisited. Automatica, 35: 1215–1241.

Gurney, W. S. C., Blythe, S. P., and Nisbet, R. M. 1980. Nicholson's blowflies revisited. Nature, 287: 17–21.

Gutiérrez, D., Sifeddine, A., Field, D. B., Ortlieb, L., Vargas, G., Chávez, F. P., Velazco, F., Ferreira, V., Tapia, P., Salvatteci, R., Boucher, H., Morales, M. C., Valdés, J., Reyss, J.-L., Campusano, A., Boussafir, M., Mandeng-Yogo, M., García, M., and Baumgartner, T. 2009. Rapid reorganization in ocean biogeochemistry off Peru towards the end of the Little Ice Age. Biogeosciences, 6: 835–848.

ICES. 2020. Inter-Benchmark Process on BAltic Sprat (Sprattus sprattus) and Herring (Clupea harengus) (IBPBash). ICES Scientific Reports. 2:34. 44 pp.

ICES. 2021. Arctic Fisheries Working Group (AFWG). ICES Scientific Reports. 3:58. 817 pp.

Lomb, N. R. 1976. Least-squares frequency analysis of unequally spaced data. Astrophysics and Space Science, 39: 447–462.

Ludwig, D., Jones, D. D., and Holling, C. S. 1978. Qualitative analysis of insect outbreak systems: the spruce budworm and forest. Journal of Animal Ecology, 47: 315–332.

McManus, M. C., Licandro, P., and Coombs, S. H. 2016. Is the Russell Cycle a true cycle? Multidecadal zooplankton and climate trends in the western English Channel. ICES Journal of Marine Science, 73: 227–238.

Moxnes, E. 1998. Not only the tragedy of the commons: misperceptions of bioeconomics. Management Science, 44: 1234–1248.

Nešić, D., and Teel, A. R. 2004. A framework for stabilisation of nonlinear sampled-data systems based on their approximate discrete-time models. IEEE Transactions on Automatic Control, 49: 1103–1122.

Pauly, D., and Tsukayama, I. (eds) 1987. The Peruvian anchoveta and its upwelling ecosystem: three decades of change. ICLARM Studies and Reviews, 15.

Punt, A. E., and Donovan, G. P. 2007. Developing management procedures that are robust to uncertainty: lessons from the International Whaling Commission. ICES Journal of Marine Science, 64: 603–612.

Ricard, D., Minto, C., Jensen, O. P., and Baum, J. K. 2012. Examining the knowledge base and status of commercially exploited marine species with the RAM Legacy Stock Assessment Database. Fish and Fisheries, 13: 380–398.

Rose, G. A. 2026. Northern cod comeback: 10 years after. Canadian Journal of Fisheries and Aquatic Sciences, 83: 1–14. https://doi.org/10.1139/cjfas-2025-0141

Rose, G. A., and Rowe, S. 2015. Northern cod comeback. Canadian Journal of Fisheries and Aquatic Sciences, 72: 1789–1798.

Rose, G. A., and Walters, C. J. 2019. The state of Canada's iconic Northern cod: a second opinion. Fisheries Research, 219: 105314.

Scargle, J. D. 1982. Studies in astronomical time series analysis. II. Statistical aspects of spectral analysis of unevenly spaced data. The Astrophysical Journal, 263: 835–853.

Statistics Canada. Tables 38-10-0167-01 and 38-10-0168-01, CANSIM database. Statistics Canada, Ottawa.

Tam, J. C., and Bundy, A. 2019. Mass-balance models of the Newfoundland and Labrador Shelf ecosystem for 1985–1987 and 2013–2015. Canadian Technical Report of Fisheries and Aquatic Sciences, 3328.

World Bank. Poverty and Inequality Platform. World Bank, Washington, DC.

Supplementary material is deposited with this article (the accompanying file `paper5_supplementary_v5.md`): the statement inventory with the status of every main-text statement (S1), the remaining proofs and identification results, including the dimensionless identifiability chart (S2), the epistemic-layer results (S3), the case-screening records at full detail (S4), the ecosystem context and toy tests (S5), the distributive-layer detail (S6), the empirical hypotheses with specified tests (S7), and the reproducibility register (S8). The register itself accompanies the article; the computational artifacts it inventories (stage registration, solver configuration, screening log, exact-update comparison) are registration requirements tracked in the register. The stage operator's boundary record is supplied by the pre-registered reconstruction of Section 3.4 — a newly specified object with archived plan, code, and outputs — while the archived stage-output values keep their provisional status until the original computational record is attached.

Declarations

Data availability

The stock and assessment series analysed in the spectral screen are drawn from the RAM Legacy Stock Assessment Database v4.66 (public release). The northern cod assessment values are published in DFO CSAS SAR 2016/026. The computational record of the response-region analyses (solver configuration, initial histories, stage registration), the exact-update comparison of Section 3.4, the RAM stock identifiers and eligibility table, the processed spectral series and routines, the power-simulation code and seeds, and the case-screening table and query log are registration requirements; the corresponding stage-output values carry provisional status until those artifacts are attached. The stage-map reconstruction of Section 3.4 is fully specified and will be deposited with the article: the dated preregistration plan (parameter table, comparison criteria), the campaign code, and the five output tables (equilibria, multiplier record, trajectory classification, robustness experiment, steepness sensitivity). Community-level social values are drawn from Statistics Canada tables 38-10-0167-01 and 38-10-0168-01 with the population-mismatch caveat stated in the text.

Author contributions

[To be completed at submission.]

Funding

[To be completed at submission.]

Conflicts of interest

The author declares no competing interests.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.